VT-Harness

An AI tutor that builds from your understanding

A visible learning process.
A memory for the next session.



Actual app with synthetic demo history

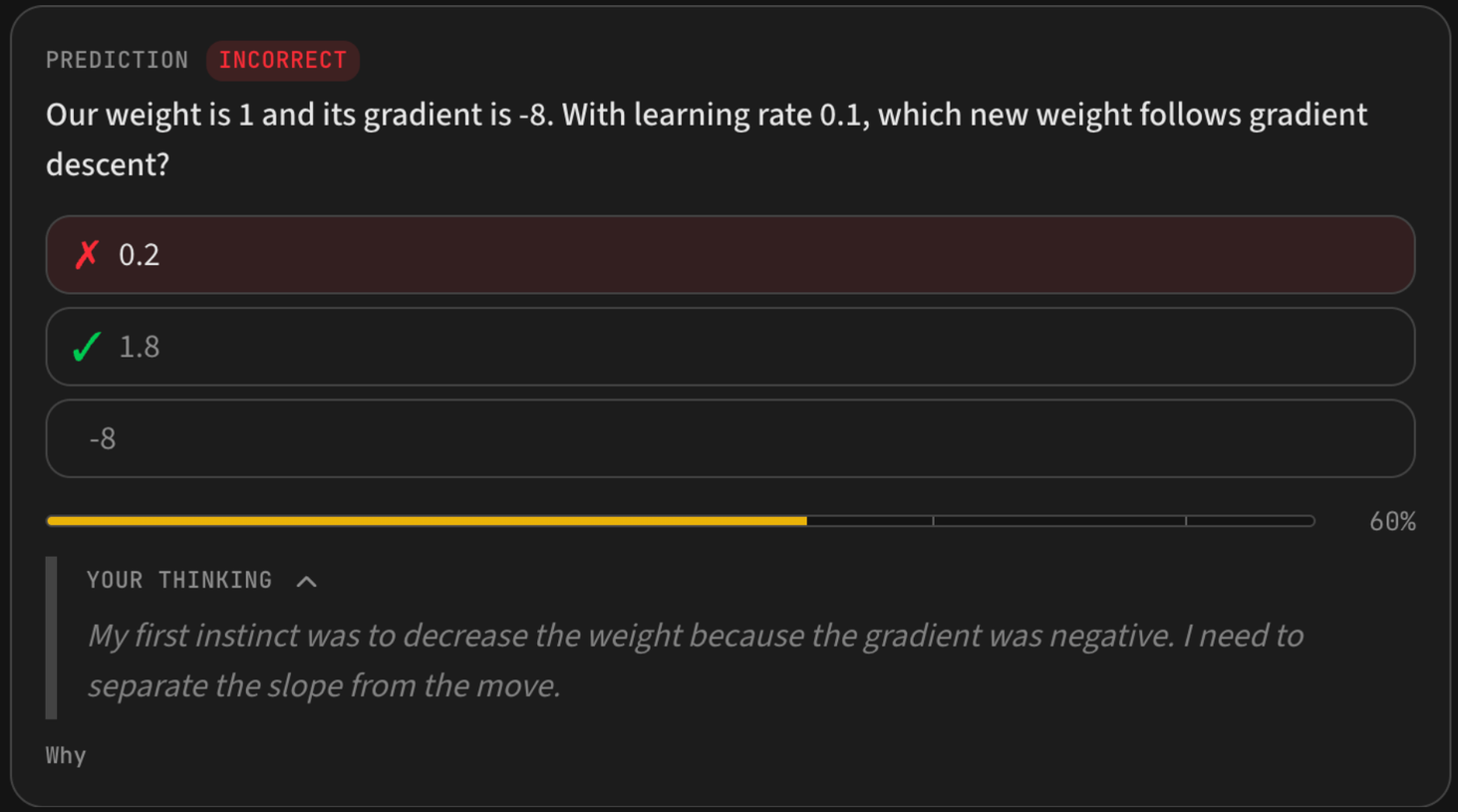
Teaching that follows your reasoning

Probe the starting point

Build a lesson plan

Predict, explain, check

Reasoning exposes the mistake.



The repair: subtracting a negative gradient increases the weight

$$1 - 0.1 \times (-8) = 1.8$$

Synthetic lesson example

Practice that tests transfer

A fresh problem

Visible reasoning

Adaptive hints

New weight: 1.9
Loss: 9 becomes 0.09

CONCLUDED SOLVED

Solved independently. You included the input factor in the gradient, subtracted the negative gradient, and checked the result by recomputing the loss.

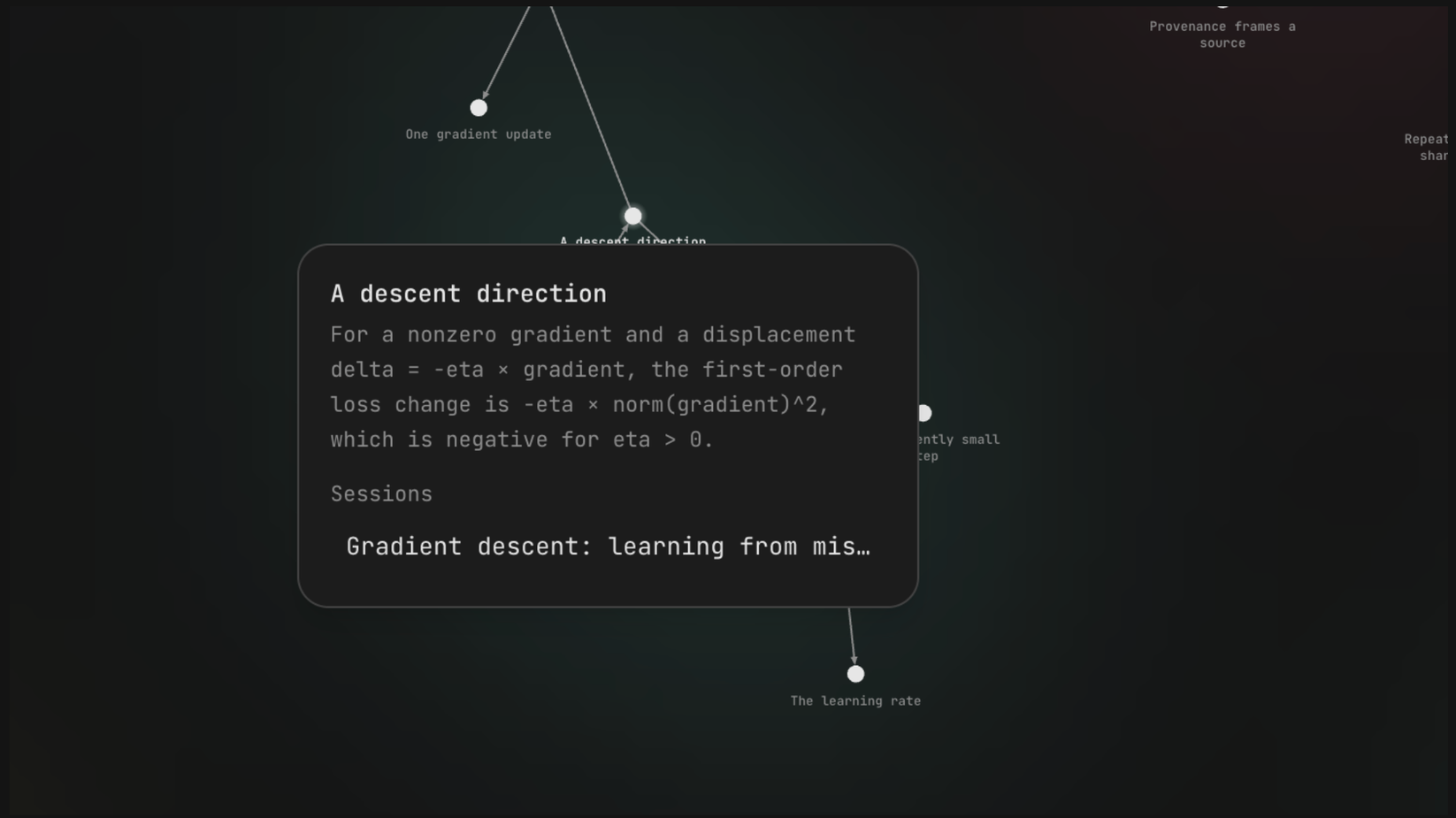
MODEL SOLUTION

Initially y_hat = 3 and the loss is (3-6)^2 = 9. The gradient is 2 × (3-6) × 3 = -18. The update is w_new = 1 - 0.05 × (-18) = 1.9. The new prediction is 5.7 and the new loss is (5.7-6)^2 = 0.09. The loss fell from 9 to 0.09.

METADATA

Synthetic practice example with a completed solution

A memory for the next session



96 concepts

53 reasoning chains

Claims and their premises, linked to the source lesson

Maths, Physics, Biology, History, Medicine and neural networks

Your material, voice and models

Your material

PDFs, files, notes and supplied URLs

Your explanations

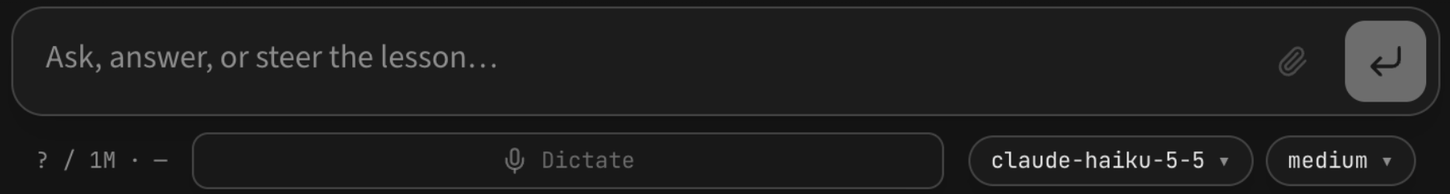
Visuals, image annotations and interactive demos

Your voice

Thinking fields and dictation

Your models

API keys, subscriptions, CLI or local endpoints



Local learning records. Open source.

github.com/ViktorIvan8787/GreatUniHack

Capability reference

Learning	Probing questions, confidence and thinking fields. Dependency plans, predictions, explanations and checks.
Practice	Graph-based problem selection, adaptive hints, learner reasoning, verdicts and model solutions.
Memory	Persistent propositions and reasoning chains. Source-session links, graph search and note import.
Materials	Supplied web URLs and PDFs, local file access, images, visual generation, annotations and interactive demos.
Control	Dictation, provider and model choice, editable prompts, context and cost visibility, local records and open source.

External providers still process the context you send. Some file and image tools require desktop.

University examples ready to explore

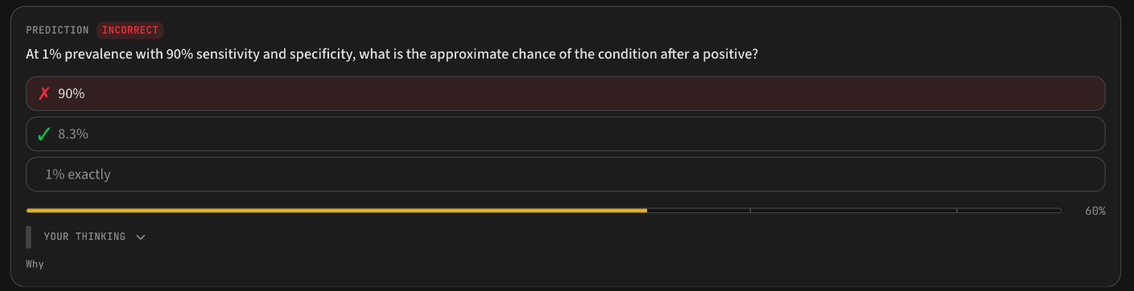
Maths
Eigenvectors and differential equations

Physics
Spring oscillations and energy

Biology
Gene regulation, selection and drift

History
The French Revolution and source criticism

Medicine
Cardiovascular physiology and diagnostic probability



Two lessons and one practice session per university discipline

Synthetic teaching history. Medicine uses fictional exercises.